EXP:4	EDA-Data Cleaning

CODE:

```
import pandas as pd


data = {
    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male',
               'Female', 'Male', 'Female', 'Male', 'Female'],
    'Married': ['Yes', 'No', 'Yes', 'Yes', 'No',
                'Yes', 'No', 'Yes', 'Yes', 'No'],
    'Dependents': [1, 0, 2, 1, '3+', 0, 2, 1, 0, 3],
    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate',
                  'Graduate', 'Not Graduate', 'Graduate', 'Graduate',
                  'Not Graduate', 'Graduate'],
    'Self_Employed': ['No', 'No', 'Yes', 'No', 'Yes',
                      'No', 'No', 'Yes', 'No', 'No'],
    'ApplicantIncome': [5000, 3000, 7000, 4500, 8000,
                        3500, 6000, 4000, 5500, 3200],
    'CoapplicantIncome': [1500, 0, 2000, 1000, 2500,
                          500, 1200, 0, 1800, 700],
    'LoanAmount': [150, 100, 200, 120, 250,
                   110, 180, 130, 160, 105],
    'Loan_Amount_Term': [360, 360, 360, 240, 360,
                         360, 180, 360, 240, 360],
    'Credit_History': [1, 1, 0, 1, 1,
```

```
                    1, 0, 1, 1, 1],
    'Property_Area': ['Urban', 'Rural', 'Semiurban', 'Urban', 'Rural',
                      'Semiurban', 'Urban', 'Rural', 'Semiurban', 'Urban']
}

df = pd.DataFrame(data)

print("First 5 Rows:")
display(df.head())

print("\nLast 5 Rows:")
display(df.tail())

print("\nDataset Information:")
df.info()

print("\nDescriptive Statistics:")
display(df.describe())

print("\nColumn Names:")
print(df.columns.tolist())

categorical_cols = [
    'Gender', 'Married', 'Dependents',
    'Education', 'Self_Employed', 'Property_Area'
]
```

```
print("\nUnique Values:")

for col in categorical_cols:
    print(f"\n{col}:")
    print(df[col].unique())

df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'], errors='coerce')

print("\nValue Counts of Dependents:")
print(df['Dependents'].value_counts())

cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']

for col in cols:
    print(f"\nStatistics for {col}:")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

OUTPUT:

First 5 Rows:

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanA
0	Male	Yes	1	Graduate	No	5000	1500	
1	Female	No	0	Graduate	No	3000	0	
2	Male	Yes	2	Not Graduate	Yes	7000	2000	
3	Female	Yes	1	Graduate	No	4500	1000	
4	Male	No	3+	Graduate	Yes	8000	2500	

Last 5 Rows:

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanA
5	Female	Yes	0	Not Graduate	No	3500	500	
6	Male	No	2	Graduate	No	6000	1200	
7	Female	Yes	1	Graduate	Yes	4000	0	
8	Male	Yes	0	Not Graduate	No	5500	1800	
9	Female	No	3	Graduate	No	3200	700	

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Gender             10 non-null     object
 1   Married            10 non-null     object
 2   Dependents         10 non-null     object
 3   Education          10 non-null     object
 4   Self_Employed      10 non-null     object
 5   ApplicantIncome    10 non-null     int64
 6   CoapplicantIncome  10 non-null     int64
 7   LoanAmount         10 non-null     int64
 8   Loan_Amount_Term   10 non-null     int64
 9   Credit_History     10 non-null     int64
 10  Property_Area      10 non-null     object
dtypes: int64(5), object(6)
memory usage: 1012.0+ bytes
```

Descriptive Statistics:

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	4970.000000	1120.000000	150.500000	318.000000	0.800000
std	1666.033213	841.691419	48.215373	69.570109	0.421637
min	3000.000000	0.000000	100.000000	180.000000	0.000000
25%	3625.000000	550.000000	112.500000	270.000000	1.000000
50%	4750.000000	1100.000000	140.000000	360.000000	1.000000
75%	5875.000000	1725.000000	175.000000	360.000000	1.000000
max	8000.000000	2500.000000	250.000000	360.000000	1.000000

```
Column Names:
['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'ApplicantIncome', 'CoapplicantI

Unique Values:

Gender:
['Male' 'Female']

Married:
['Yes' 'No']

Dependents:
[1 0 2 '3+' 3]

Education:
['Graduate' 'Not Graduate']

Self_Employed:
['No' 'Yes']

Property_Area:
['Urban' 'Rural' 'Semiurban']

Value Counts of Dependents:
Dependents
1    3
0    3
2    2
3    2
Name: count, dtype: int64

Statistics for ApplicantIncome:
Mean: 4970.00
```

```
Statistics for ApplicantIncome:
Mean: 4970.00
Median: 4750.00
Mode: [3000 3200 3500 4000 4500 5000 5500 6000 7000 8000]
Range: 5000.00
Variance: 2775666.67
Standard Deviation: 1666.03

Statistics for CoapplicantIncome:
Mean: 1120.00
Median: 1100.00
Mode: [0]
Range: 2500.00
Variance: 708444.44
Standard Deviation: 841.69

Statistics for LoanAmount:
Mean: 150.50
Median: 140.00
Mode: [100 105 110 120 130 150 160 180 200 250]
Range: 150.00
Variance: 2324.72
Standard Deviation: 48.22
/tmp/ipykernel_11634/2807943612.py:56: FutureWarning: Downcasting behavior in `replace` is deprec
  df['Dependents'] = df['Dependents'].replace('3+', 3)
```